Quizrr Chapter-Wise Test for JEE Advanced - 2025

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Q.1

$S = \{1, 2, 3, 5, 8, 13, 21, 34\}$. Find $\sum \max(A)$ where the sum is taken over all 28 two elements subsets A to S.

(A) 844

(B) 480

(C) 484

(D) 488

Q.2

If the variate of a distribution takes the values $1^2, 2^2, 3^2, \ldots n^2$ with frequencies $n, n-1, n-2, \ldots 3,2,1$ respectively, then the mean value of the distribution is

(A) $\frac{n(n+2)}{3}$

(B) $\frac{n(n+1)(n+2)}{6}$

(C) $\frac{n+2}{3}$

(D) $\frac{(n+1)(n+2)}{6}$

Q.3

The mean and the standard deviation (s.d.) of 10 observations are 20 and 2 respectively. Each of these 10 observations is multiplied by p and then reduced by q, where $p \neq 0$ and $q \neq 0$. If the new mean and new s.d. become half of their original values, then q is equal to

(A) -5

(B) 10

(C) -20

(D) -10

Q.4

Find median of following frequency distribution :

Class interval	$10-25$	$25-40$	$40-55$	$55-70$	$70-85$	$85-100$
$\mathbf{f_i}$	6	20	44	26	3	1

(A) 44.54

(B) 46.62

(C) 48.18

(D) 50.36

Q.5

The age distribution of workers in a factory is as follows:

Age in Years	**No. of Workers**

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$20-28$	45
$28-36$	95
$36-44$	100
$44-52$	42
$52-60$	18

If 15% of the total strength starting from the lowest age group is retrenched and 20% of the total strength from the highest age groups is given premature retirement, then the range of age of workers retained in the factory is

(A) $20-36$

(B) $28-44$

(C) $28-52$

(D) $36-52$

Q.6

Let $G=\{(b,b),(b,c),(c,c),(c,d)\}$ and $H=\{(b,a),(c,b),(d,c)\}$. Then the number of elements in the set $(G\cup H)\oplus(G\cup H)^{-1}$, where $\oplus$ denotes the symmetric difference, is

(A) 0

(B) 2

(C) 7

(D) 14

Q.7

The actual mode of the data set below

Misplaced \hline

(A) 11.50

(B) 10.50

(C) 10.25

(D) 11.25

Q.8

If $\alpha,\ \beta,\ \gamma$ are the number of elements in sets A, B and C respectively, and sum of number of all subsets of A, B and C is 28, evaluate the maximum number of ordered triplets $(\alpha,\ \beta,\ \gamma)$ that can be formed.

(A) 1

(B) 2

(C) 4

(D) 6

Q.9

Let $f(x)=x^2,\ x\in R$. For any $A\subseteq R$, define $g(A)=\{x\in R\ :f(x)\in A\}$. If $S=[0,\ 4]$, then which one of the following statements is true?

(A) $g(f(S))\neq S$

(B) $f(g(S))\neq f(S)$

(C) $f(g(S))=S$

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(D) $g(f(S)) = g(S)$

Q.10

For $2n$ observations $a_1, -a_1, a_2, -a_2, a_3, -a_3, ..., a_n, -a_n$ where all a's are distinct. Let $\overline{x}$, σ and M denotes mean, standard deviation and median, respectively. Then which of the following options are correct ?

(A) $\overline{x} = 0$

(B) $\overline{x} = M$

(C) $\sigma > \overline{x}$

(D) $\sigma < M$

Q.11

Let W denote the words in the English dictionary. Define the relation R by : $R = \{(x, y) \in W \times W |$ the words x and y have at least one letter in common $\}$.
Then R is

(A) reflexive relation

(B) symmetric relation

(C) transitive relation

(D) equivalence relation

Q.12

There is a raffle with 12 tickets and two prizes of Rs. 9 and Rs. 6. A, B, C whose probabilities of speaking the truth are $\frac{1}{2}, \frac{2}{3}, \frac{3}{5}$ respectively, report the result to D, whose holds one ticket. A and B assert that the has won the Rs. 9 prize, and C asserts that he has won the Rs. 6 prize; what is D's expectation?

(A) less than 3

(B) greater than 1.75

(C) less than 2

(D) greater than 1.25

Q.13

Let R be the real line. Consider the following subsets of the plane $R \times R$

$S = \{(x,\ y) : y = x + 1 \text{ and } 0 < x < 2\}$

$T = \{(x,\ y) : x - y \text{ is an integer}\}$

Which of the following is true?

(A) S and T are equivalence relation

(B) S is symmetric and T is equivalence relation

(C) S is not symmetric and T is transitive relation

(D) S is not symmetric and T is equivalence relation

Q.14

A relation R is defined as $(x,\ y) \in R \Rightarrow x^y = y^x$ for $x,\ y \in I - \{0\}$, where I is the set of all integers. Then the relation R is:

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(A) reflexive relation

(B) symmetric relation

(C) transitive relation

(D) equivalence relation

Q.15

Paragraph

A bag contains 6 balls. Two balls are drawn from this bag and both are found to be white. Let random variable x, be the number of white balls in the bag. Now answer the following, given that the two drawn balls are kept back in the bag.**Question**

If initially it is equally likely that bag contains any number of white balls then after this drawing of two balls and their replacement expected number of white balls in the bag is

(A) 4

(B) $\frac{9}{2}$

(C) 5

(D) $\frac{34}{7}$

Q.16

Paragraph

A bag contains 6 balls. Two balls are drawn from this bag and both are found to be white. Let random variable x, be the number of white balls in the bag. Now answer the following, given that the two drawn balls are kept back in the bag.**Question**

If initially it is equally likely that a ball is white or black then after this drawing of two balls and their replacement expected number of white balls in the bag is

(A) 4

(B) $\frac{9}{2}$

(C) 5

(D) $\frac{34}{7}$

Q.17

Given a number n, find out the number of Symmetric Relations on a set of first n natural numbers $\{1, 2, ..n\}$. (Take value of n=8)

(A) 2^{32}

(B) 2^{36}

(C) 2^{40}

(D) 2^{28}

Q.18

Given a number n, find out the number of Reflexive Relation on a set of first n natural numbers $\{1, 2, ..n\}$. (Take value of n=6)

(A) 2^{30}

(B) 2^{76}

(C) 2^{100}

(D) 2^{90}

Q.19

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Suppose $A_1,\ A_2,\ \ldots,\ A_{30}$ are thirty sets each having 5 elements and $B_1,\ B_2,\ \ldots.,\ B_n$ are n sets each with 3 elements. Let $\bigcup_{i=1}^{30} A_i = \bigcup_{j=1}^{n} B_j = S$ and each element of S belongs to exactly 10 of the $A_i's$ and exactly 9 of the $Bj's$. Then n is equal to

Q.20

The algebraic sum of the deviation of 20 observation measured from 30 is 2. Then the mean of observation is

Answers & Solutions

Q.1 Answer:

484

Solution:

$S = \{1, 2, 3, 5, 8, 13, 21, 34\}$

All the subset of two elements of set S are

$\{1,2\},\ \{1,3\},\ \{1,5\},\ \{1,8\},\ \{1,13\},\ \{1,21\},\ \{1,34\},\ \{2,3\},\ \{2,5\},\ \{2,8\},\ \{2,13\},\ \{2,21\},\ \{2,34\},\ \{3,5\},\ \{3,8\},\ \{3,13\},\ \{3$

$\therefore \sum \max(A)$

$= \begin{bmatrix} 2+3+5+8+13+21+34+3+5+8+13+21+34+5+8+13+21+34+8+13 \\ +21+34+13+21+34+21+34+34 \end{bmatrix}$

$= 2 + 2(3) + 3(5) + 4(8) + 5(13) + 6(21) + 7(34)$

$= 2 + 6 + 15 + 32 + 65 + 126 + 238$

$= 484$

Q.2 Answer:

$\frac{(n+1)(n+2)}{6}$

Solution:

$x_i \ : \ 1^2 \quad 2^2 \quad 3^2 \quad 4^2 \quad \ldots \quad (n-1)^2 \quad n^2$

$f_i \ : \ n \quad n-1 \quad n-2 \quad n-3 \quad \ldots \quad 2 \quad 1$

$\therefore \sum_{i=1}^{n} x_i f_i = 1^2 \cdot n + 2^2(n-1) + 3^2(n-2) + \ldots + (n-1)^2 x + n^2 \cdot 1$

$= \sum_{r=1}^{n} (n+1)r^2 - r^3$

$= (n+1)\sum_{r=1}^{n} r^2 - \sum_{r=1}^{n} r^3$

$= \frac{n(n+1)^2(2n+1)}{6} - \frac{n^2(n+1)^2}{4}$

$\frac{n(n+1)^2}{12}\{2(2n+1) - 3n\} = \frac{n(n+1)^2(n+2)}{12}$

Also, $\Sigma f_i = 1 + 2 + \ldots + n = \frac{n(n+1)}{2}$

Mean $= \frac{\Sigma f_i x_i}{\Sigma f_i} = \frac{n(n+1)^2(n+2)}{12} \times \frac{2}{n(n+1)}$

$= \frac{(n+1)(n+2)}{6}$

Q.3 Answer:

-20

Solution:

If each observation is multiplied with p & then q is subtracted

New mean $\bar{x}_1 = p\bar{x} - q$

$\Rightarrow\ 10 = p(20) - q \ldots\ldots (1)$

and new standard deviations

$\sigma_2 = |p|\sigma_1 \ \Rightarrow\ 1 = |p|(2) \ \ \Rightarrow\ \ |p| = \frac{1}{2} \ \ \Rightarrow\ p = \pm\frac{1}{2}$

If $p = \frac{1}{2}$

then $q\ =\ 0$ **(from equation (1))**

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If $p = -\frac{1}{2}$

$q = -20$

Q.4 Answer:

48.18

Solution:

Cumulative frequency table

Class interval	$\mathbf{f_i}$	**Cumulative frequency (c)**
$10 - 25$	6	6
$25 - 40$	20	26
$40 - 55$	44	70
$55 - 70$	26	96
$70 - 85$	3	99
$85 - 100$	1	100

$N = 100$

Here, $\frac{\mathrm{N}}{2} = 50 \Rightarrow$ Class interval of median is "40 - 55"

Here, $\ell = 40,\ c = 26,\ h = 15,\ f = 44$

$$\therefore \quad \text{The median (M)} = \ell + \frac{\left(\frac{\mathrm{N}}{2} - \mathrm{c}\right)}{\mathrm{f}} \times \mathrm{h} = 40 + \frac{(50 - 26)}{44} \times \mathrm{h} = 40 + \frac{24}{44} \times 15 = 48.18$$

Therefore the median is 48.18

Q.5 Answer:

$28 - 44$

Solution:

From the given table, the total number of workers $= 300$

Retrenched $= 15\%$ **of** $300 = 45$

These are all from age group $(20 - 28)$**. (Given)**

Premature retired $= 20\%$ **of** $300 = 60$

So, 18 **workers from age group** $(52 - 60)$ **and** 42 **workers from age group** $(44 - 52)$**.**

$\therefore$ **The age limit of workers retained is** $28 - 44$**.**

Q.6 Answer:

2

Solution:

$G = \{(b, b), (b, c), (c, c), (c, d)\}$

$H = \{(b, a), (c, b), (d, c)\}$

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$(G \cup H) = \{(b,b),(b,c),(c,c),(c,d),(b,a),(c,b),(d,c)\}$

$(G \cup H)^{-1} = \{(b,b),(c,b),(c,c),(d,c),(a,b),(b,c),(c,d)\}$

$(G \cup H) \oplus (G \cup H)^{-1}$

$= \left[(G \cup H) - (G \cup H)^{-1}\right] \cup \left[(G \cup H)^{-1} - (G \cup H)\right]$

$= \{(b,a)\} \cup \{(a,b)\}$

$= \{(b,a),(a,b)\}$

Hence, number of elements in

$(G \cup H) \oplus (G \cup H)^{-1}$ **is** 2.

Q.7 Answer:

11.25

Solution:

$\text{Mode} = \text{L} + \left(\frac{\text{f}_1 - \text{f}_0}{2\text{f}_1 - \text{f}_0 - \text{f}_2}\right) \times \text{h}$

Where

- **L = 10**
- **f_1 = 9**
- **f_0 = 4**
- **f_2 = 2**
- **h = 3**

Therefore,

$\text{Mode} = 10 + \left(\frac{9-4}{(2\times 9)-2-4}\right) \times 3$

$\text{Mode} = 10 + \left(\frac{5}{12}\right) \times 3$

$\text{Mode} = 10 + 1.25$

$\text{Mode} = 11.25$

Q.8 Answer:

6

Solution:

We know that the number of subsets of a set containing n elements is 2^n.
Hence, the number of subsets of $A,\ B$ and C are respectively $2^\alpha,\ 2^\beta$ and 2^γ.

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Given, $2^{\alpha}+2^{\beta}+2^{\gamma}=28$

$\Rightarrow\ \alpha,\ \beta,\ \gamma$ **can be** $(2,\ 3,\ 4),\ (2,\ 4,\ 3),\ (3,\ 2,\ 4),\ (3,\ 4,\ 2),\ (4,\ 2,\ 3)$ **and** $(4,\ 3,\ 2)$

Hence, total 6 **possibilities are there.**

Q.9 Answer:

$g(f(S))\neq S$

Solution:

From the given information we can conclude

$g(S)=[-2,2]$ **, So,** $f(g(S))=[0,\ 4]=S$

And $f(S)=[0,\ 16]\Rightarrow f(g(S))\neq f(S)$

Also, $g(f(S))=[-4,\ 4]\neq g(S)$

So, $g(f(S))\neq S$

Hence, $g(f(S))=g(S)$ **is not true.**

Q.10 Answer:

$\overline{x}=0$

Solution:

Mean, $\overline{x}=\frac{a_1+(-a_1)+a_2+(-a_2)+(a_3)+(-a_3)+.....+a_n+(-a_n)}{2n}=0$

If n **is odd, then in the series of** $2n$ **observation middle terms are** $a_{\frac{n+1}{2}}\ \&\ -a_{\frac{n+1}{2}}$

$\therefore\ \ M=0$

Standard deviation, $\sigma=\sqrt{\frac{(a_1)^2+(-a_1)^2+(a_2)^2+(-a_2)^2+(a_3)^2+(-a_3)^2+...+(a_n)^2+(-a_n)^2}{2n}-(\overline{x})^2}$

$\sigma=\sqrt{\frac{a_1^2+a_1^2+a_3^2+...+a_n^2}{n}-0}=\sqrt{\frac{a_1^2+a_2^2+a_3^2+...+a_n^2}{n}}$

Q.11 Answer:

reflexive relation

Solution:

Given relation R **such that**

$R=\{(x,y)\in W\times W|$ **the world** x **and** y **have atleast one letter in common}**

Where W **denotes set of words in English dictionary Clearly**

$(x,x)\in R\,\forall\,x\in W$

$\therefore\ \ (x,x)$ **has every letter common** $\therefore R$ **is reflexive Let** $(\mathrm{x},\mathrm{y})\in R$ **then**

$(y,x)\in R$**as** x **and** y **have atleast one letter in common.** $\Rightarrow R$ **is symmetric.**

But R **is not transitive**

$\therefore$ **Let** $x=$ **DON,** $y=$ **NEST,** $z=$ **SHE**

Then $(x,y)\in R$ **and** $(y,z)\in R$**. But** $(x,z)\notin R$ **.**

$\therefore\ R$ **is reflexive, symmetric but not transitive.**

Q.12 Answer:

less than 3

Solution:

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Three cases are possible; D may have won Rs. 9, Rs. 6, or nothing, for A, B, C may all have spoken falsely.

Now we have the a priori probabilities

$$\mathrm{P}_1 = \frac{1}{12}, \mathrm{P}_2 = \frac{1}{12}, \mathrm{P}_3 = \frac{10}{12};$$

$$\text{also } \mathrm{p}_1 = \frac{1}{2} \times \frac{2}{3} \times \frac{2}{5} = \frac{4}{30}, \mathrm{p}_2 = \frac{1}{2} \times \frac{1}{3} \times \frac{3}{5} = \frac{3}{30},$$

$$\mathrm{p}_3 = \frac{1}{2} \times \frac{1}{3} \times \frac{2}{5} = \frac{2}{30};$$

$$\frac{\mathrm{P}_1\mathrm{p}_1}{4} = \frac{\mathrm{P}_2\mathrm{p}_2}{3} = \frac{\mathrm{P}_3\mathrm{p}_3}{20} = \frac{1}{27};$$

hence D's expectation

$= \frac{4}{27}$ of Rs. $9 + \frac{3}{27}$ of Rs. $6 =$ Rs. 2

Q.13 Answer:

S is not symmetric and T is transitive relation

Solution:

Since, $(1,\ 2) \in S$ **but** $(2,\ 1) \notin S$

$\therefore S$ **is not symmetric.**

Hence, S **is not an equivalent relation.**

Given, $T = \{(x,\ y) : (x - y) \in I\}$

Now, $xTx \Rightarrow x - x = 0 \in I$**, it is reflexive relation**

Again, $xTy \Rightarrow (x - y) \in I$

$\Rightarrow y - x \in I \Rightarrow yTx$ **it is symmetric relation.**

Let xTy **and** yTz

$\therefore x - y = I_1$ **and** $y - z = I_2$

Now, $x - z = (x - y) + (y - z) = I_1 + I_2 \in I$

$\Rightarrow x - z \in I$

$\Rightarrow \quad xTz$

$\therefore T$ **is transitive.**

Hence, T **is an equivalence relation.**

Q.14 Answer:

reflexive relation

Solution:

$\because (x,\ y) \in R \Rightarrow x^y = y^x$

$\therefore (x,\ x) \in R$ **as** $x^x = x^x\ \forall\ x \in I - \{0\}$

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$\therefore R$ **is reflexive**

Now $(x,\ y) \in R \Rightarrow x^y = y^x \Rightarrow y^x = x^y$

$\Rightarrow (y,\ x) \in R \quad \therefore R$ **is symmetric**

Now $\quad (x,\ y) \in R \Rightarrow x^y = y^x$ **and** $(y,\ z) \in R \Rightarrow y^z = z^y$

$\because x^y = y^x \Rightarrow y = x^{\frac{y}{x}}$

$\Rightarrow y^z = z^x \Rightarrow \left(x^{\frac{y}{x}}\right)^z = z^x \Rightarrow x^{\frac{yz}{x}} = z^x$

$\Rightarrow x^{yz} = z^{x^2} \Rightarrow \left(x,\ z\right) \notin R$

R **is not transitive.**

Q.15 Answer:

5

Solution:

$\frac{{}^6C_2}{{}^6C_2} \times 6 + \frac{{}^5C_2}{{}^6C_2} \times 5 + \frac{{}^4C_2}{{}^6C_2} \times 4 + \frac{{}^3C_2}{{}^6C_2} \times 3 + \frac{{}^2C_2}{{}^6C_2} \times 2 = \frac{90+50+24+9+2}{15} \times \frac{1}{7} = \frac{175}{15\times 7}$

${}^2C_2 + {}^3C_2 + {}^5C_2 + {}^6C_2 = 35$

given two white $P(x=2) = \frac{\frac{1}{7} \times \frac{{}^2C_2}{{}^6C_2}}{\frac{1}{7}\left(\frac{35}{{}^6C_2}\right)} = \frac{1}{35}$

Q.16 Answer:

4

Solution:

$$P(X = 2/\text{two white balls}) = \frac{\frac{{}^6C_2}{2^6} \times \frac{{}^2C_2}{{}^6C_2}}{\frac{{}^6C_2}{2^6} \times \frac{{}^2C_2}{{}^6C_2} + \frac{{}^6C_3}{2^6} \times \frac{{}^3C_2}{{}^6C_2} + \frac{{}^6C_4}{2^6} \times \frac{{}^4C_2}{{}^6C_2} + \frac{{}^6C_5}{2^6} \times \frac{{}^5C_2}{{}^6C_2} + \frac{{}^6C_2}{2^6} \times \frac{{}^6C_2}{{}^6C_2}}$$

$P(X=2) = \frac{{}^6C_2 \times {}^2C_2}{15+60+90+60+15}$

$= \frac{1}{240}\left(15 \times 2 + 60 \times 3 + 90 \times 4 + 60 \times 5 + 15 \times 6\right)$

$= \frac{960}{240}$

Q.17 Answer:

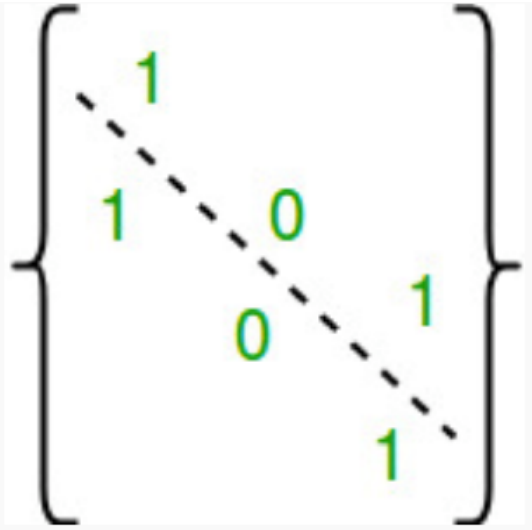
2^{36}

Solution:

Total number of symmetric relations is $2^{n(n+1)/2}$.

How does this formula work?

A relation R **is symmetric if the value of every cell** (i, j) **is same as that cell** (j, i)**. The diagonals can have any value.**



There are n **diagonal values, total possible combination of diagonal values** $= 2^n$

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There are $n^2 - n$ non-diagonal values. We can only choose different value for half of them, because when we choose a value for cell (i, j), cell (j, i) gets same value.

So combination of non-diagonal values $= 2^{(n2-n)/2}$

Overall combination $= 2^{n*}2^{(n2-n)/2} = 2^{n(n+1)/2}$

Q.18 Answer:

2^{30}

Solution:

The number of reflexive relations on an n-element set is $2^{n(n-1)}$ How does this formula work?

A relation R is reflexive if the matrix diagonal elements are 1 .

$$\begin{Bmatrix} 1 & & & \\ & 1 & & \\ & & 1 & \\ & & & 1 \end{Bmatrix}$$

If we take a closer look the matrix, we can notice that the size of matrix is n^2. The n diagonal entries are fixed.

For remaining $n^2 - n$ entries, we have choice to either fill 0 or 1.

So there are total $2^{\mathrm{n(n-1)}}$ ways of filling the matrix.

Q.19 Answer:

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Solution:

$\sum_{i=1}^{30} n(Ai) = 5 \times 30 = 150$

Let S consist of m distinct elements. Since each elements of S belongs to exactly 10 of the Ai', so, we have

$\sum_{i=1}^{30} n(A_i) = 10m = 150 \Rightarrow m = 15$

Now each B_i has 3 elements and each elements of S belongs to exactly 9 of the B_i's. So, we have

$\sum_{j=1}^{n} n(B_j) = 3n = 9m \Rightarrow 3n = 9 \times 15 \Rightarrow n = 45$

Q.20 Answer:

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Solution:

We have, $\sum_{i=1}^{20}(x_i - 30) = 2$

Therefore, $(x_1 + x_2 + \ldots\ldots\ldots + x_{20}) = 602$

Now, $\overline{x} = \frac{x_1+x_2+\ldots\ldots+x_{20}}{20}$

$= \frac{602}{20}$

$= 30.1$

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